

Data Analytics and Visualization App

app development with data analytics and visualization

Comprehensive Specification Book

System Architecture

Our system architecture consists of three main layers:

Presentation Layer

- Frontend: Built using React.js for a responsive and user-friendly interface
- API Gateway: Handles incoming requests and routes them to the appropriate service

Business Logic Layer

- Data Analytics Service: Utilizes libraries like pandas and NumPy for data manipulation and analysis
- Data Visualization Service: Leverages libraries like D3.js and Chart.js for interactive visualizations

Data Storage Layer

- Database: Utilizes a relational database like PostgreSQL for storing and retrieving data
- Data Warehouse: Utilizes a cloud-based data warehouse like Amazon Redshift for storing and processing large datasets

Use Case Diagram

```
graph LR
    A[User] -->|Login|> B[Authentication Service]
    B -->|Validate|> C[Database]
    C -->|Return Token|> A
    A -->|Request Data|> D[Data Analytics Service]
    D -->|Process Data|> E[Data Warehouse]
    E -->|Return Data|> A
    A -->|Visualize Data|> F[Data Visualization Service]
    F -->|Return Visualization|> A
```

Sequence Diagram for Main User Flow

```
sequenceDiagram
    participant User as A
    participant Authentication Service as B
    participant Database as C
    participant Data Analytics Service as D
    participant Data Warehouse as E
    participant Data Visualization Service as F

    A->>B: Login
    B->>C: Validate
    C->>B: Return Token
    B->>A: Return Token
    A->>D: Request Data
    D->>E: Process Data
    E->>D: Return Data
    D->>A: Return Data
    A->>F: Visualize Data
    F->>A: Return Visualization
```